

ECON 212 — Principles of Microeconomics

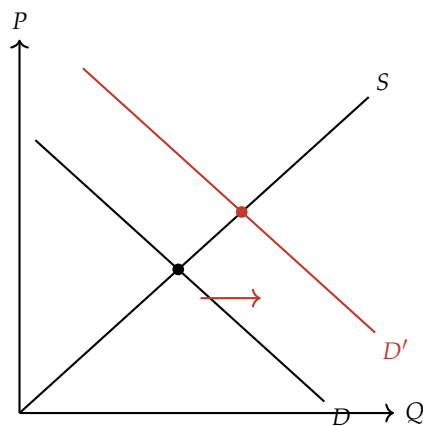
Lecture Notes — Ch. 5: Elasticity

Fall 2026

Picks up where Supply & Demand left off: shifters and their magnitudes.

→ We are quantifying things.

Remember...



Q What happens to P? To Q?

→ $P \uparrow, Q \uparrow$.

But here is the killer question:

Q Which one rises more? Or how much of a change will occur?

- * How should we make this comparison? **Q** (open question)
 - ↪ Using different units! \$ vs. # widgets.
- * Depends on the shapes/slopes of the curves.
 - ↪ We call this “elasticity”.

Outline

How this chapter is laid out:

- ① Define elasticity.
- ② Understand the nuances and how it relates to the different kinds of elasticities.
- ③ Show the power of elasticity to understand economic problems.

① What Is *an* Elasticity?

A measure of how much Q_D or Q_S changes when a determinant changes.

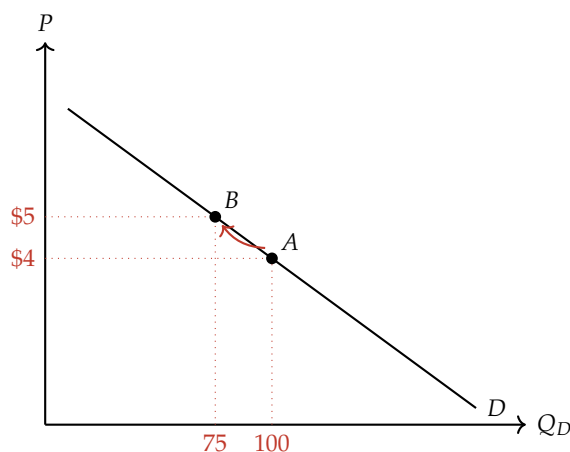
→ Price is one determinant, but not the only one!

“Price elasticity of demand”: how much Q_D changes when price changes.

→ Q_D is measured in #; price is measured in \$.

⇒ Put them both as percents.

Picture



(This looks like slope...)

First, find how Q_D changes.

$$\text{Change in } Q_D = \Delta Q_D = \underbrace{Q_D^B - Q_D^A}_{75 - 100} = -25$$

$$\% \text{ change in } Q_D = \Delta Q_D \% = \frac{\Delta Q_D}{Q_D^A} = \frac{-25}{100} = \boxed{-25\%}$$

where Q_D^A is the “first” Q_D — the one we started from.

Second, find how P changes.

$$\text{Change in } P = \Delta P = P_B - P_A = \$5 - \$4 = \$1$$

$$\Delta P \% = \frac{\Delta P}{P_A} = \frac{\$1}{\$4} = \boxed{25\%}$$

→ So a 25% increase in P leads to a 25% decrease in Q_D !

How do we put these together?

Price elasticity of demand

this is “epsilon”

$$\downarrow \epsilon_P = \frac{\Delta Q_D\%}{\Delta P\%} = \frac{-25\%}{25\%} = -1 !$$

Suppose $\epsilon_P = \frac{1}{5}$, so a 10% increase in P leads to a 2% decrease in Q_D .

Going the Other Way — and Why It Breaks

Great work! Now, for more practice, find the price elasticity of demand going the other way:

$$\begin{aligned} \Delta Q_D\% &= \frac{25}{75} = 33\% \\ \Delta P\% &= \frac{4-5}{5} = -20\% \\ \Rightarrow \epsilon_P &= \frac{33\%}{-20\%} = -1.65 \dots \end{aligned}$$

Note: by convention, we always report price elasticity as a positive number, even though it is always negative.

Ah shucks, this isn't right???

So it turns out that going $A \rightarrow B$ AND $B \rightarrow A$ give different answers...

Why is this?

$$\epsilon_P(A \rightarrow B) = \frac{\left(\frac{\Delta Q_D}{Q_D^A}\right)}{\left(\frac{\Delta P}{P_A}\right)} \neq \frac{\left(\frac{\Delta Q_D}{Q_D^B}\right)}{\left(\frac{\Delta P}{P_B}\right)} = \epsilon_P(B \rightarrow A)$$

The Midpoint Method

How do we resolve this difference?

Instead of Q_D^A or Q_D^B , take their average.

$$\begin{aligned} \text{Average}(Q_D^A, Q_D^B) &= \frac{Q_D^A + Q_D^B}{2} = 87.5 \\ \text{Average}(P_A, P_B) &= \frac{P_A + P_B}{2} = 4.5 \end{aligned}$$

The midpoint method

$$\varepsilon_P(A \rightarrow B) = \varepsilon_P(B \rightarrow A) = \frac{\left[\frac{Q_D^B - Q_D^A}{\left(\frac{Q_D^A + Q_D^B}{2} \right)} \right]}{\left[\frac{P_B - P_A}{\left(\frac{P_A + P_B}{2} \right)} \right]}$$

Give in-class practice

Critical: use A, B, C to show that elasticity changes along the curve

Big note: elasticity changes along the curve. Why?

→ Because the midpoint changes.

Terminology: Elastic and Inelastic

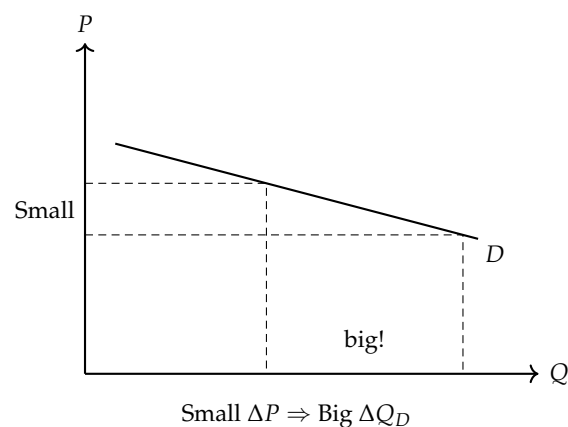
Good work. Now, let's add some terminology here.

Use this as a time to reiterate what an elasticity means!

Elastic

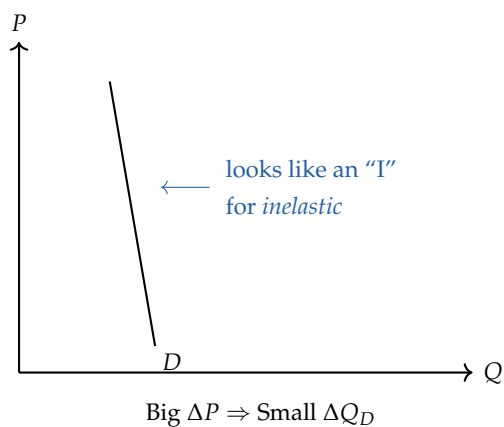
Q_D responds *substantially* to P : small $\Delta P \Rightarrow$ big ΔQ_D , so $\varepsilon_P > 1$.

Q How would this look?



Inelastic

Q_D responds *slightly* to P : big $\Delta P \Rightarrow$ small ΔQ_D , so $\varepsilon_P < 1$.



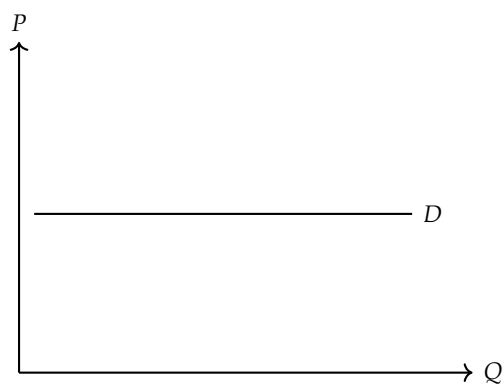
Edge Cases

Let's work through possible edge cases.

Q Ask for ideas.

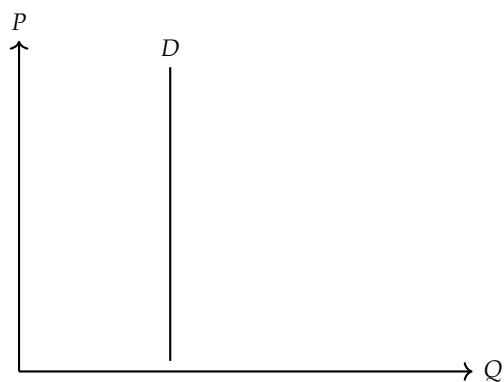
Perfectly elastic

Any change in price $\Rightarrow \infty$ change in Q_D , so $\epsilon_P = \infty$.



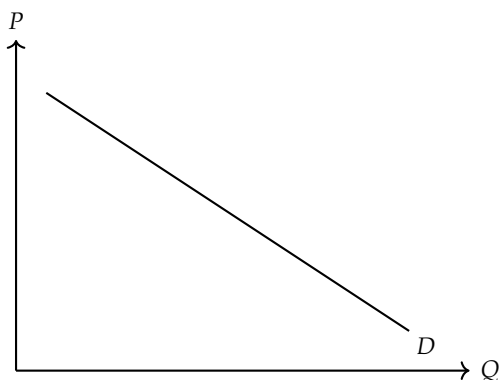
Perfectly inelastic

Any change in price $\Rightarrow 0$ change in Q_D , so $\epsilon_P = 0$.



Unit elastic

A 1% change in price $\Rightarrow$ a 1% change in Q_D .



② The Nuances — Where Elasticities Come From, and What Other Kinds There Are

We now know what an elasticity is and how to measure one. But why is demand for some goods very elastic, and for others very inelastic? In other words, where do elasticities come from?

Q What are some things that are very elastic or inelastic?

→ Use this to intro each category

① Availability of Close Substitutes

Q Examples?

Close	None
Oat milk & almond milk	Eggs (flax eggs?)
Chicken & turkey	Cars (bikes)

② How the Market Is Defined (Narrow vs. Broad)

→ Cars → *inelastic*

↔ Honda Civic ↔ *elastic*

↔ Toyota Corolla ↔ *elastic*

Narrow definitions allow for more close substitutes.

→ Meat vs. chicken, beef, pork...

③ Luxuries vs. Necessities

Q Examples?

Lux	Nec
Yacht	Food
Filet mignon	Water
Rolex	Housing
Sports cars	Health care
NFL tickets	Clothing
Elastic	Inelastic

④ Time Horizon

Longer time frame $\Rightarrow$ more elastic.

Q Examples?

$\rightarrow$ Ex: gas.

$\hookrightarrow$ High price *today* $\Rightarrow$ no change.

$\hookrightarrow$ High price *for a year* $\Rightarrow$ hybrid, EV, carpool, live closer to work...

In-Class Exercise: Ranking the Elasticities

Rank the following by elasticity, from most inelastic to most elastic.

Break into teams

Good	$ \varepsilon_P $	Good	$ \varepsilon_P $	Good	$ \varepsilon_P $
Spirits (hard liquor)	0.8	Gasoline (short run)	0.1	Gasoline (long run)	0.3
Cereal	0.6	Milk	0.6	Cigarettes	0.4
Health care	0.2	Soda	16.6	Wine	0.7
Eggs	0.3	Yogurt	4.4	Toilet paper	3.6
Potato chips	4.1	Potatoes	6.3	Bread (buns, fresh)	12.8
Tortilla chips	7.7	Salad dressing	4.8	Electricity (SR)	0.23
Fruit Loops	3.5	Air travel	0.8	Spaghetti	3.0
Fruit	0.7	Vegetables	0.6	Crackers	5.5
Sweets / sugars	0.3	Sugar-free gum	5.3	Granola bars	5.7
Frozen pizza	6.2	Beer	0.5	Spaghetti sauce	4.2
Fresh meat	3.2				

What Other Kinds of Elasticities Are There?

Q What are the shifters of the demand curve?

- * Tastes / preferences *can't quantify...*
- * Expectations *can't quantify...*
- * Income ★
- * Prices of related goods ★
- * # buyers *mechanical / unit elastic*

① Income Elasticity of Demand

Income elasticity of demand

$$\varepsilon_I = \frac{\Delta Q_D\%}{\Delta I\%} = \frac{\text{percentage change in } Q_D}{\text{percentage change in income}}$$

Q What key term or type of good from ECON 211 might be related to ε_I ?

→ Normal & inferior goods.

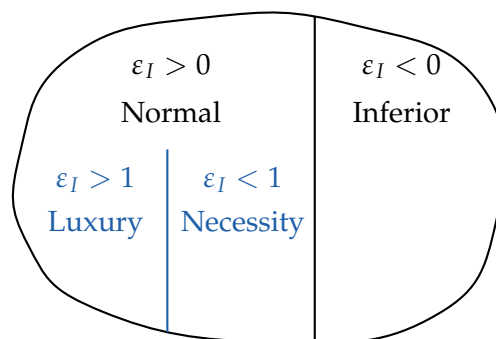
Normal $\varepsilon_I > 0 \Rightarrow$ the more income, the more you buy.

Q Why might some goods have high or low ε_I ?

→ Necessity: low ε_I , b/c you are already buying it.

→ Luxury: high ε_I [$\varepsilon_I > 1$].

Inferior $\varepsilon_I < 0 \Rightarrow$ more income, buy less.



Note

Income elasticities are always reported with their sign.

② Cross-Price Elasticity of Demand

How much Q_D changes when the price of a particular related good changes.

Cross-price elasticity of demand

$$\varepsilon_{CP} = \frac{\Delta Q_D\%}{\Delta P_{OG}\%} \quad \text{where } OG \text{ is the other good.}$$

Q What would it mean if the cross-price elasticity were positive?

→ The goods are substitutes! $\varepsilon_{CP} > 0 \Rightarrow$ substitutes.

Negative?

→ Complements!

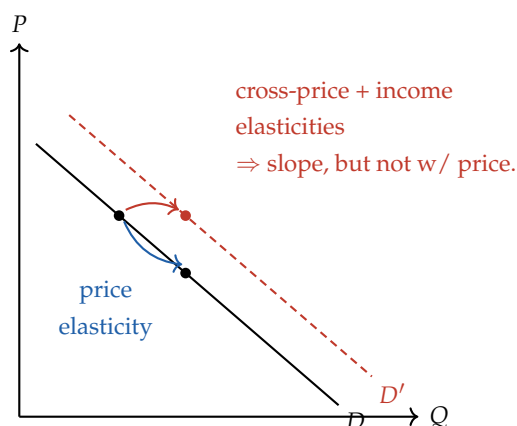
⇒ Now we have a technical/quantitative way to understand something we knew in words before!

* This is powerful. We used math to understand an old concept in a new way.

Welcome to theory

A Helpful Picture

Perhaps a helpful picture for understanding elasticities.



One Other Elasticity: Supply

If we did demand, we have to do supply too...

Price elasticity of supply

How much Q_S changes as price changes.

$$\varepsilon_S = \frac{\Delta Q_S\%}{\Delta P\%}$$

...And everything else is the same...

→ With the exception of the determinants of supply elasticity!

↔ The main difference is whether the good is limited in quantity.

+ Beachfront land on Lake Macatawa? Very limited.

+ Toothpaste? Easy to produce more, quickly.

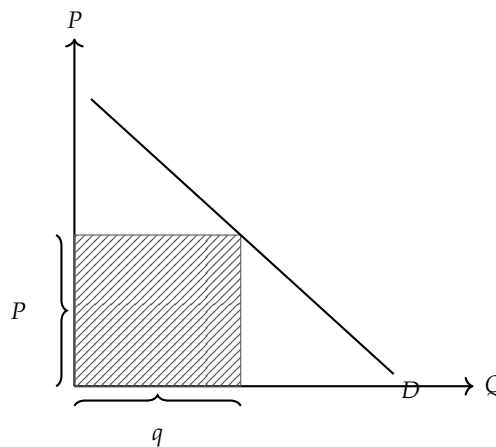
③ The Power of Elasticity

Let's use the elasticity now to understand a few things.

① Total Revenue

0 b/c it is out of order from the book

This is the total number of \$ paid for the good. In math, it is $P \times Q$. In pictures:

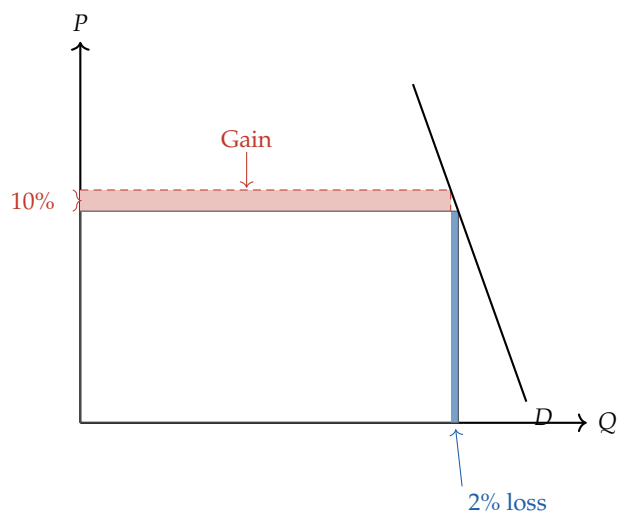


You have probably laid awake many nights wondering “how do I maximize total revenue?” Today you will learn about how elasticities can be used to do this!

Let's Do This in Pictures

A Inelastic demand: what happens if price *increases*?

Inelastic $\Rightarrow \epsilon_P < 1 \Rightarrow$ small change in Q_D .



What happens to total revenue?

$$P \rightarrow 1.1 P$$

$$Q \rightarrow (1 - 0.02) Q = 0.98 Q$$

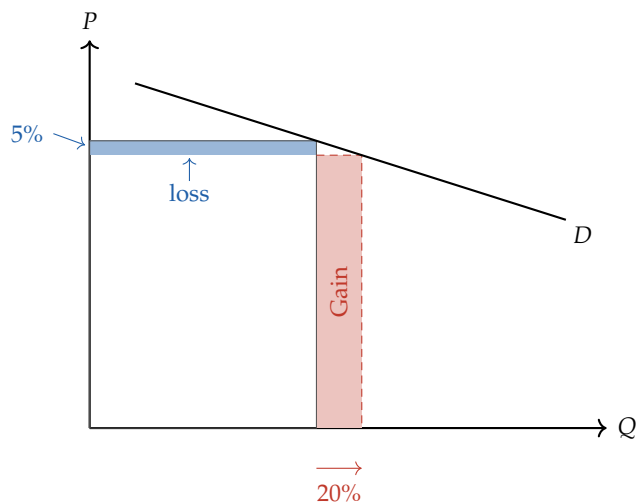
$$PQ \rightarrow 1.1 \cdot P \cdot 0.98 \cdot Q = 1.08 PQ > PQ \quad \checkmark$$

The 10% gain in P is relatively larger than the 2% loss in Q !

B Elastic demand: what happens if price decreases?

Elastic $\Rightarrow \varepsilon_P > 1 \Rightarrow$ large change in Q_D .

Suppose $\varepsilon_P = 4$, so a 5% decrease in P leads to a 20% increase in Q_D .



Again,

$$\begin{aligned}
 P &\longrightarrow 0.95 P \\
 Q_D &\longrightarrow 1.2 Q \\
 \Rightarrow 1.2 \times 0.95 PQ &= 1.14 PQ > PQ
 \end{aligned}$$

Relatively larger gain in Q .

Q So where is revenue maximized?

→ Where there is unit elasticity!

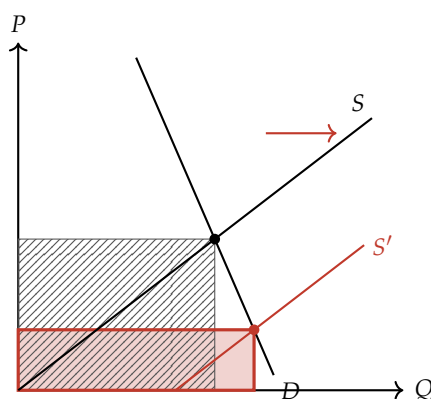
Use blind mountain climber example from 312 here

① How Do Supply Shifts Affect Seller Welfare?

→ A farmer in Kansas. If new tech increases productivity, will you be better off?

Q Price elasticity of demand for wheat: elastic or inelastic?

Not many substitutes, large market definition



So total farming revenue falls, b/c D is inelastic!

Q Can this help explain a few things?

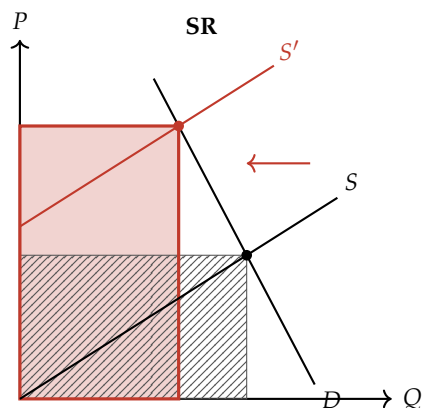
→ As tech increases, farmers have a lower opportunity cost to selling the farm and moving to the city.

→ Why the government would pay farmers to leave land idle.

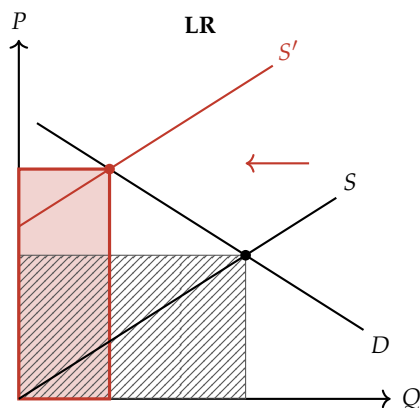
Q So why do they adopt this technology?

② Why Can't Suppliers Keep Prices High?

→ OPEC in the 70s and 80s reduced production to raise the price of oil.



In the short run this is very beneficial, b/c demand is inelastic.



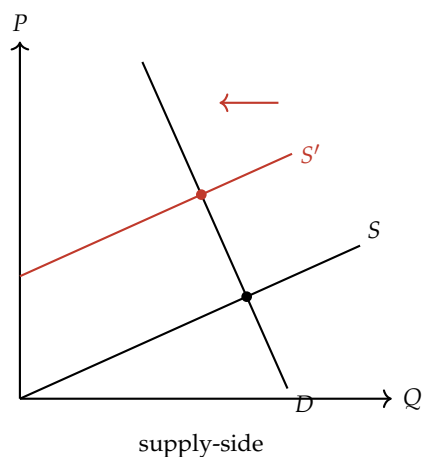
In the long run demand is elastic.

→ Alternatives exist!

③ Should Demand or Supply Be Targeted?

→ Why has the war on drugs been so difficult?

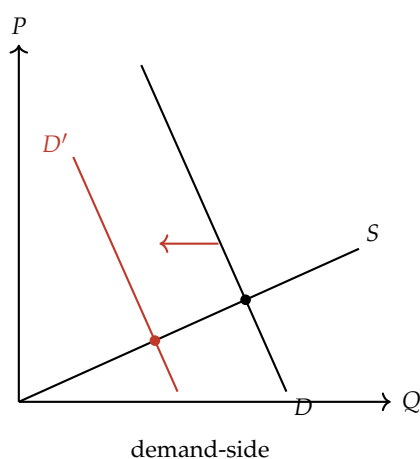
↪ Drug demand is inelastic.



Arresting smugglers, increasing penalties $\Rightarrow$ S shift.

What happens to total revenue? It goes up $\uparrow$ Q? Barely moves.

Crimes related to stealing \$ to buy drugs? $\uparrow$



Drug education?

Less revenue *and* lower price!

→ Fewer crimes.